High-Level Synthesis with Game Theoretic Scheduling

[M4 — Part 4 of 4: Advantages, Applications, Future Work, Conclusion]

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I. Advantages and Limitations

The proposed game theoretic technique has numerous key advantages for power optimization in High-Level Synthesis. Operations are extracted from the Data Flow Graph (DFG) and each functional unit is treated as a game player. Each participant decides the operation it can perform with the least power expenditure. The Nash equilibrium is then used to obtain the assignment that minimizes the power consumption rate at the functional unit for each operation. This results in decreased switching activity at the functional units and thus in minimization of the overall power consumption.

Another major advantage is that this strategy does not increase the hardware area. No new adders, multipliers or other functional units are added to the design. Instead, the existing hardware components are used more efficiently through better scheduling and binding. Thus, the hardware area is basically unchanged and only the assignment of operations to units is optimized.

However, the strategy also has certain limitations. The proposed method prioritizes minimizing power consumption rather than optimizing latency, so sometimes additional control cycles may be required, which may cause a slight increase in delay [1].

II. Applications and Relevance

This strategy is very beneficial in IoT devices, wearables and mobile devices because the battery in these devices is small and the power constraint is highly strict. Less power means longer battery life, less heat and better efficiency of the device.

HLS makes hardware design easier. Game theory and Nash equilibrium can be used to allocate operations to functional units with lower power consumption. That makes scheduling and binding decisions more routine.

In the future, if AI or machine learning is applied, it can learn from prior design expertise. This allows for more intelligent judgments to be made regarding which hardware unit consumes less power, where latency can be reduced and how calculation time can be minimized for larger circuits [1].

III. Future Work

In this work, the main focus is on reducing power consumption. In practical hardware design, however, low power and good speed are equally vital. If simply power is reduced, the circuit may be slow. Therefore, the extension of this approach to jointly optimize power and latency is an interesting topic for future exploration.

In addition to this, interconnect binding should be taken into account in the method. In current chips, the power consumption and delay are not only from functional units but also from wires and interconnects. But the optimization is not complete without the interconnects consideration.

The computation of the Nash equilibrium requires much more time for large circuits with many operations and hardware components. To solve this, further study can examine quicker algorithms, heuristic approaches, search space reduction methods or AI and machine learning based ways to minimize the computation time [1].

IV. Conclusion

This work has demonstrated that the scheduling and binding problem in High-Level Synthesis can be effectively solved using Nash equilibrium, and that power consumption can be reduced through this approach. The main strength of this method is that it assigns operations to suitable functional units in a way that minimizes power without increasing the hardware area.

However, further work still remains. Future improvements should focus on reducing latency alongside power, incorporating interconnect binding into the framework, and making the Nash equilibrium calculation faster for large and complex circuits [1].

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